Nerves of the Hand

- IS-MSN 2024
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Objectives: Upon completion of this lecture, you should be able to discuss and explain:

1) The course and distribution of median nerve and its branches to the muscles and skin of the hand.
2) The predicted motor and sensory deficits with trauma to a) median nerve at the elbow and b) median nerve at the wrist.
3) The course and distribution of ulnar nerve and its branches to the muscles and skin of the hand.
4) The predicted motor and sensory deficits with trauma a) to ulnar nerve at the elbow, and b) entrapment of ulnar nerve in the Canal of Guyon.
5) The course and distribution of radial nerve in the forearm and hand.

Review: Begin by reviewing the course of median and ulnar nerves in the anterior forearm, and radial nerve in the posterior forearm. Refer back to FOM directed studies and dissecting guides as needed.

Note that the **ulnar nerve** at the elbow courses posteriorly in the ulnar nerve groove (cubital tunnel) of the medial epicondyle, a site of potential compression and ulnar nerve palsy. Recall that as ulnar nerve enters the anterior forearm, it runs medially to innervate 1½ muscles: flexor carpi ulnaris (FCU) and the ulnar half of flexor digitorum profundus (FDP).

Review the relationships of **median nerve** to the tendon of the biceps brachii and the brachial artery as median nerve crosses the cubital fossa. Recall that median nerve enters the forearm first through the two heads of pronator teres, and then deep to the sublimis bridge formed by the expansive origin of the flexor digitorum superficialis (FDS). Median nerve innervates all of the muscles of the anterior forearm except the 1½ muscles served by ulnar nerve as discussed above.

Remember that median nerve gives off the **anterior interosseous nerve** in the proximal forearm and this branch supplies only 3 muscles deep in the anterior forearm: flexor pollicis longus, lateral half of FDP, and pronator quadratus. Anterior interosseous n. terminates in pronator quadratus and *never enters the hand*, while median nerve proper travels through the carpal tunnel to enter the hand.

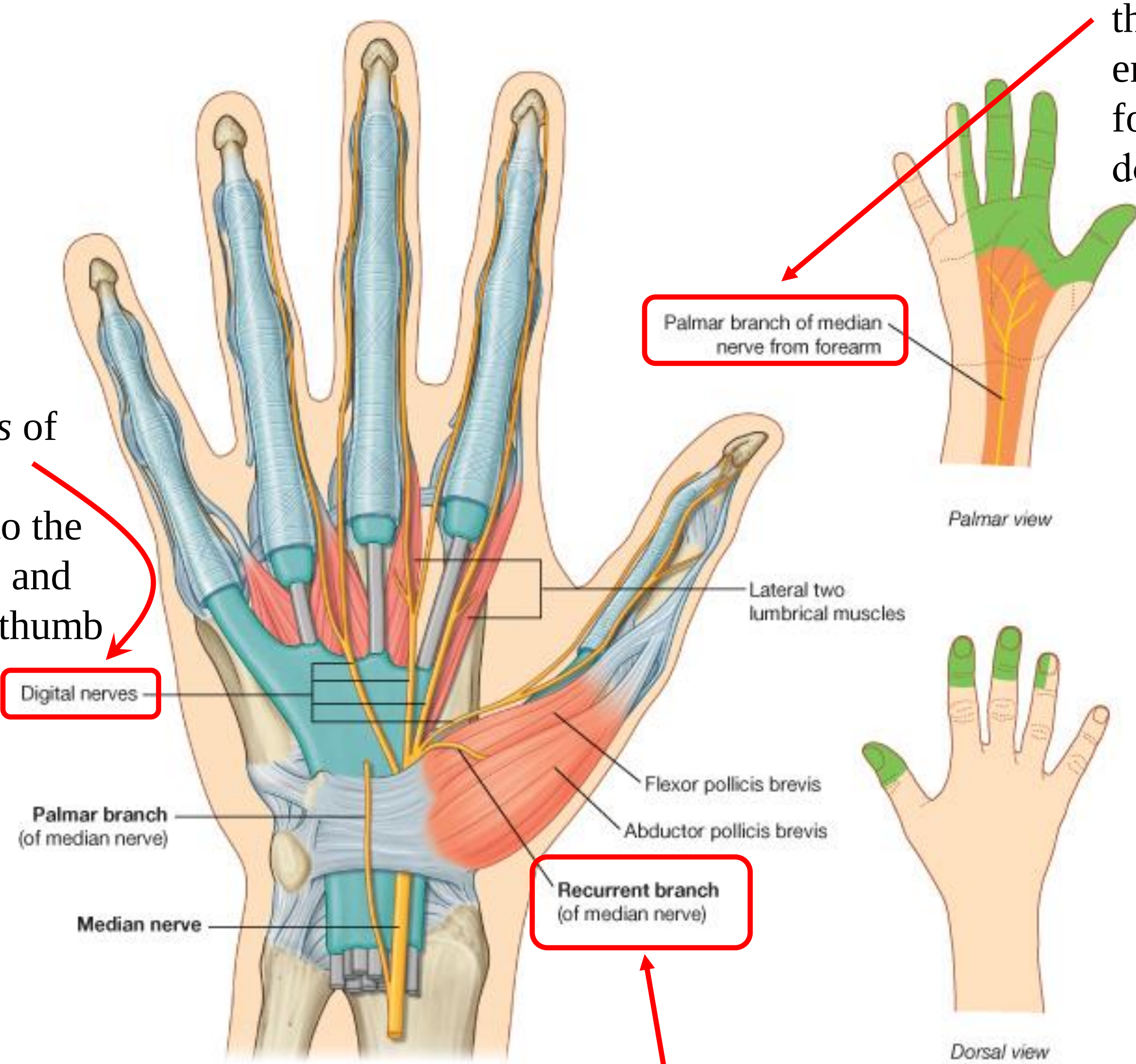
Radial nerve courses anterior to the lateral epicondyle and capitulum at the elbow, and then emerges between the brachioradialis and brachialis muscles to divide into two branches: a **deep branch** and a **superficial branch**.

Radial n. proper innervates brachioradialis and extensor carpi radialis longus. The **deep branch** supplies extensor carpi radialis brevis and supinator, and upon exiting supinator, becomes the **posterior interosseous n**., which innervates the remaining forearm extensor muscles. Posterior interosseous n. terminates at the dorsum of the wrist, which it supplies. Thus, **no motor branch of radial nerve crosses the wrist.**

Superficial radial n. is a *sensory* branch that courses anterolaterally, under cover of the brachioradialis muscle in the forearm. Superficial radial nerve then continues into the hand, running superficially over the anatomical snuffbox. This nerve supplies some cutaneous branches to the skin of the hand to some of the joints of the hand but is does not supply any of the intrinsic muscles of the hand. It is strictly a sensory nerve to skin and joints of the hand.

Median nerve in the hand

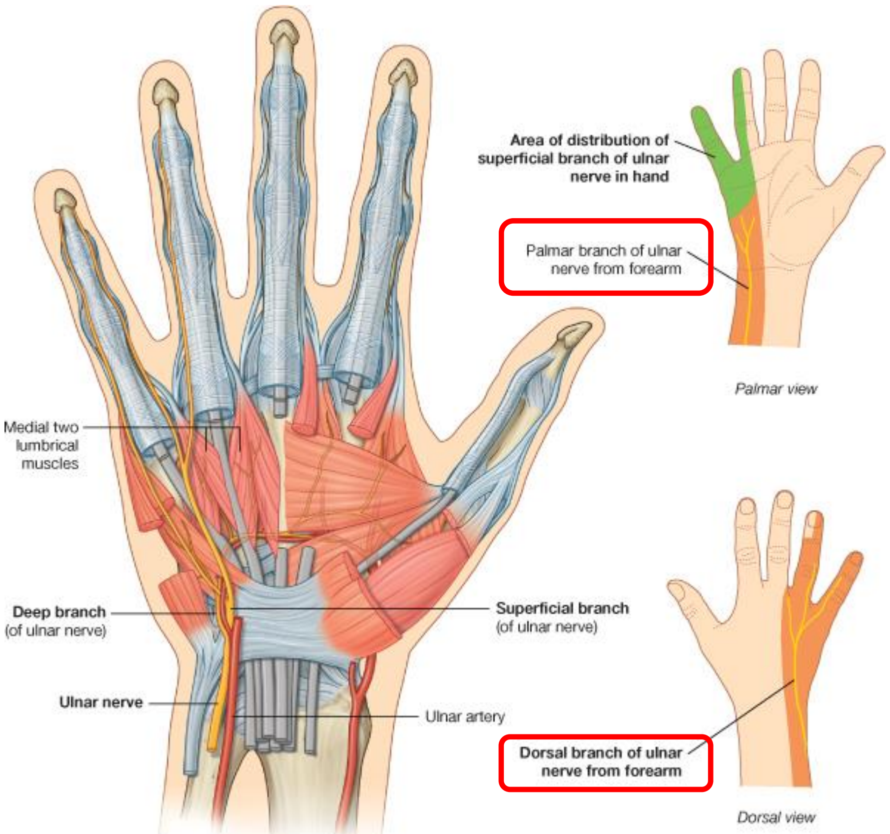
the *palmar branch* emerges in the distal forearm and thus doesn't pass through the carpal tunnel

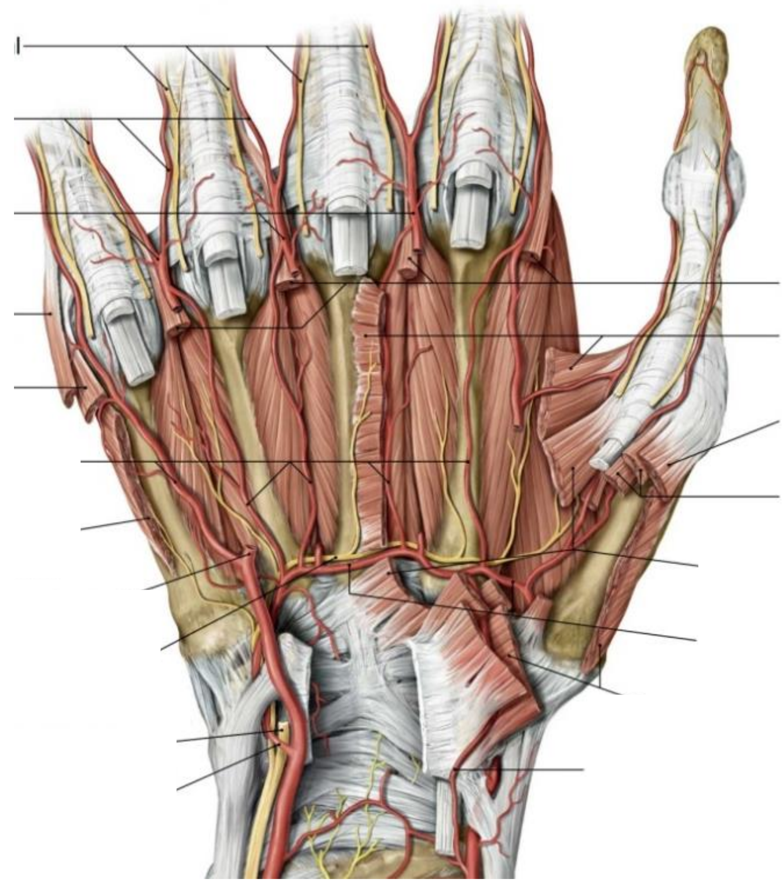


the *digital branches* of the median provide motor innervation to the lateral 2 lumbricals and also sensory to the thumb and the lateral 2.5 fingers (& their dorsal tips)

the *recurrent branch* of the median supplies the thenar mm.

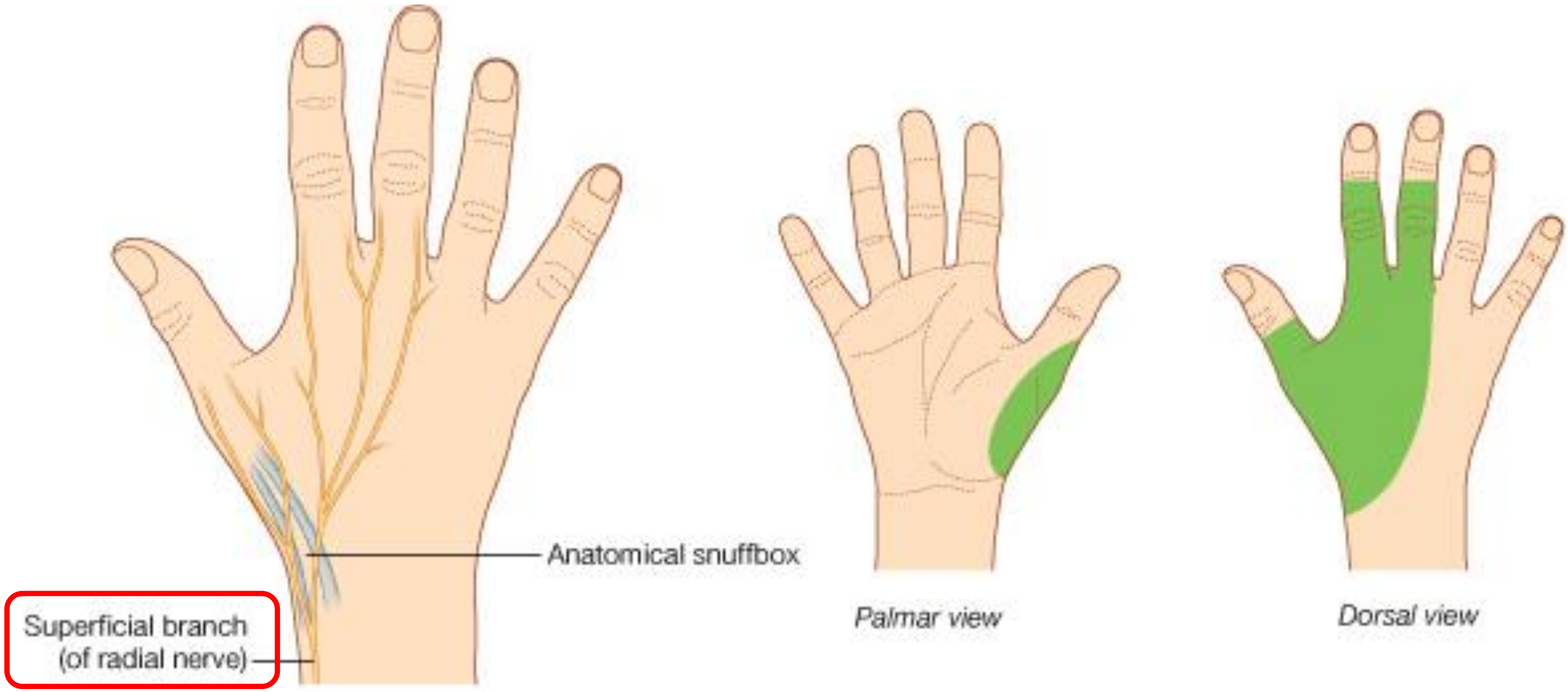
Ulnar nerve in the hand





the *palmar* and *dorsal sensory branches* arise proximal to the wrist and won't be compressed in the Canal of Guyon

Radial nerve in the hand:



superficial branch of the radial nerve passes superficial to the anatomical snuffbox and then provides sensory innervation to portions ot the dorsum of the hand

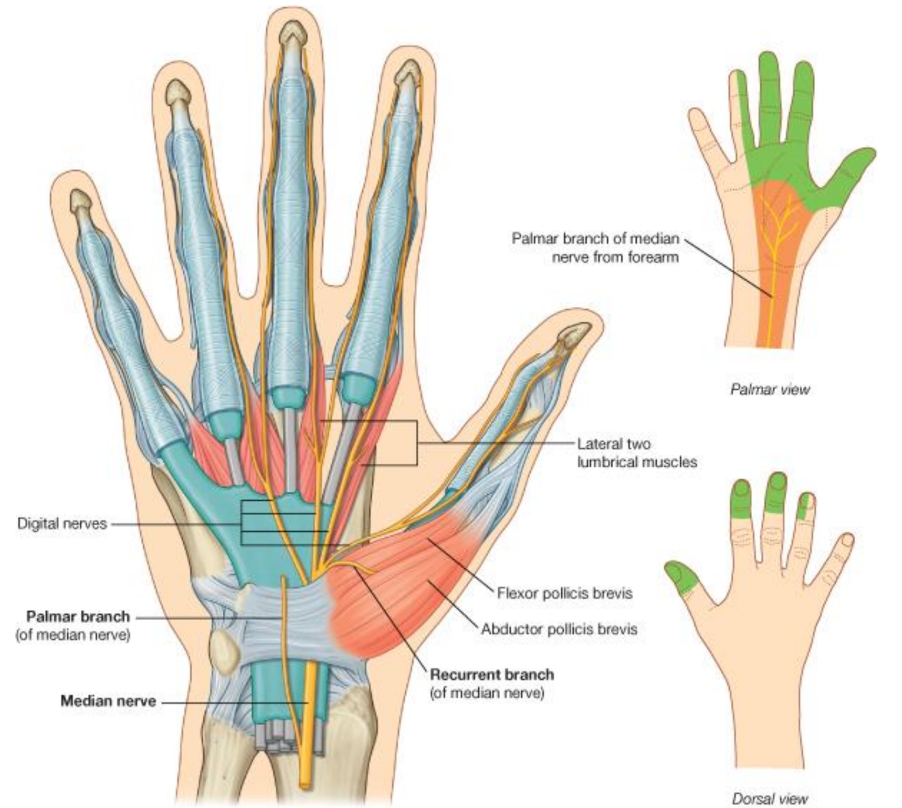
Median nerve in the hand:

Median nerve runs through the **carpal tunnel**, superficial to the synovial sheath that houses the long digital flexor tendons and deep to the flexor retinaculum. The carpal tunnel is a common site of entrapment of median n.

Upon entering the hand, median nerve immediately gives rise to the ***recurrent branch of median nerve***, which supplies the three thenar muscles. The recurrent branch of median nerve terminates in opponens pollicis.

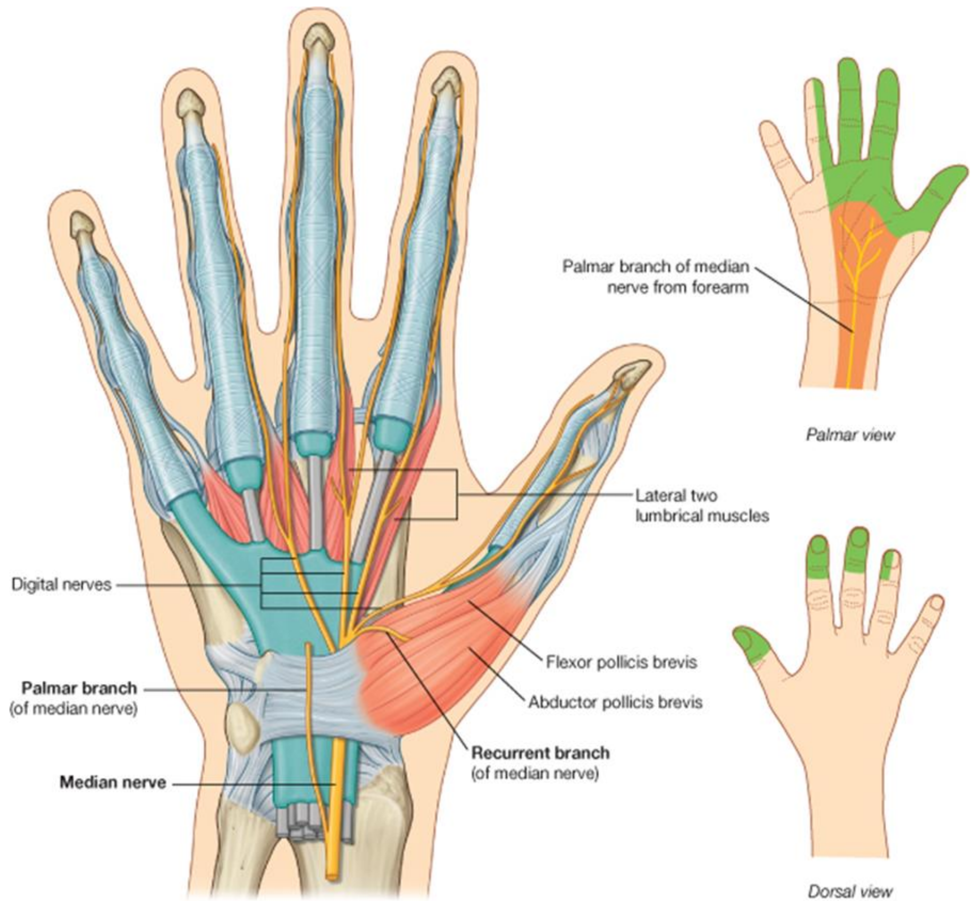
Median nerve proper proceeds to give rise to **4 digital nerves**: two of these innervate the lateral two lumbrical muscles, and then all four distribute cutaneous innervation to the distal palmar skin, the skin of the palmar thumb, and the skin of the palmar lateral 2½ digits, extending to the dorsal fingertips.

Again, notice that the skin of the distal anterior forearm and proximal had is innervated by a cutaneous branch of median nerve (**palmar branch**) before this nerve enters the carpal tunnel.



Injury to the Median Nerve in the Carpal Tunnel:

mainly accompanied by loss of sensation on lateral 3+ digits and inability to oppose the thumb; however, because the lateral lumbricals are also knocked out, digits 2 and 3 will also flex more slowly than digits 4 and 5.



Injury to the Median Nerve in the Forearm:

- forearm held in supination (because pronators are knocked out)
- wrist flexion weak and accompanied by adduction (as FCR is knocked out),
- no IP flexion of digits 2 and 3 because FDP to those digits is out
- thus ***trying to make a fist*** yields no flexion of digits 1-3, and weaker/partial flexion of digits 4-5 (due to loss of FDS), thumb adducted. This is what is often referred to as the **“hand of benediction”**

Ulnar nerve in the hand

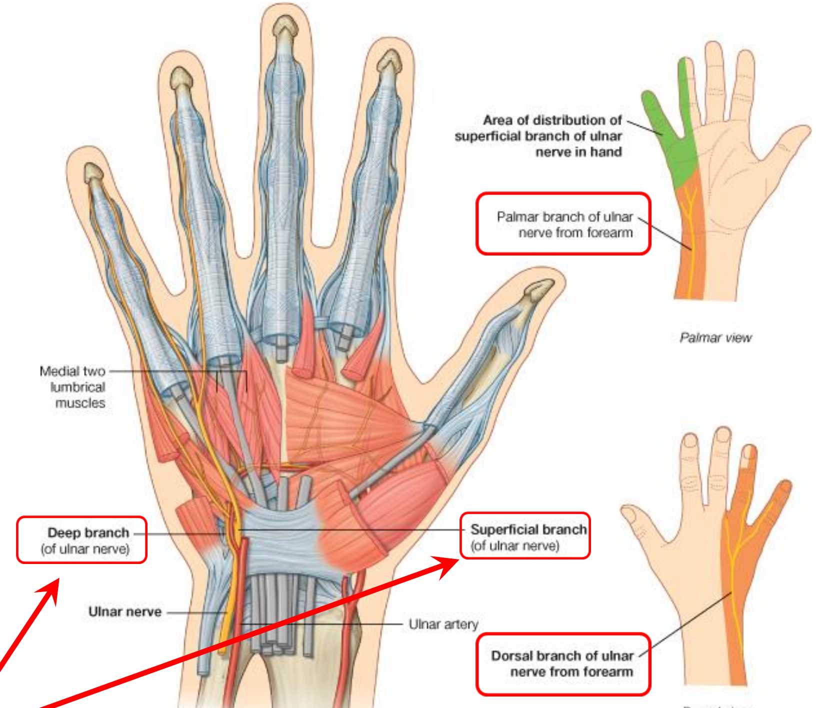
In the distal forearm, ulnar nerve gives rise to palmar and dorsal cutaneous branches.

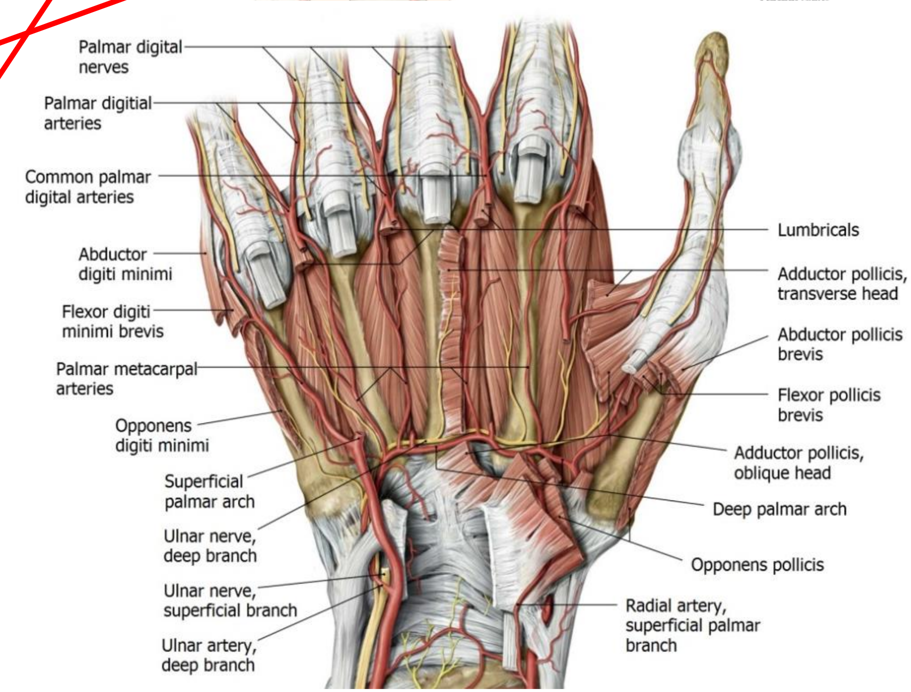
- **Palmar branch**: skin on the anterior medial side of the wrist and palm
- **Dorsal branch**: skin on the dorsomedial side of the wrist and hand and dorsum of medial 1½–2 fingers (dorsal branch).

When ulnar n. enters the hand, it passes through the **Canal of Guyon** (see next slide) and then gives off two branches:

–***superficial branch***: supplies palmaris brevis (inconsequential muscle, not shown here), and then divides into digital nerves to supply the palmar skin of the medial 1½ digits and associated distal palmar skin

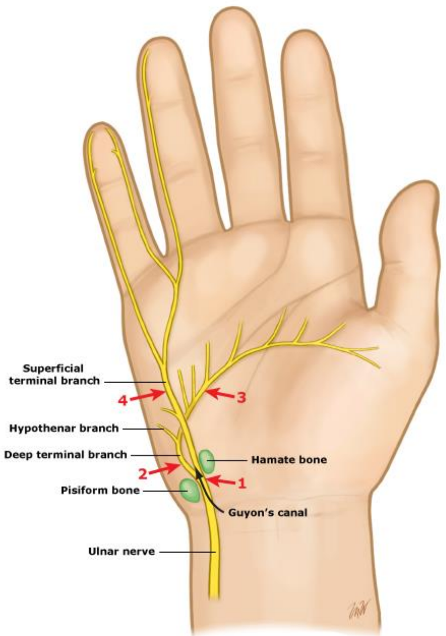
–***deep branch***: supplies the hypothenar muscles, then dives deep and arches laterally across the deep surface of the palm to supply the two medial lumbricals, all of the palmar and dorsal interossei, and the adductor pollicis muscle.

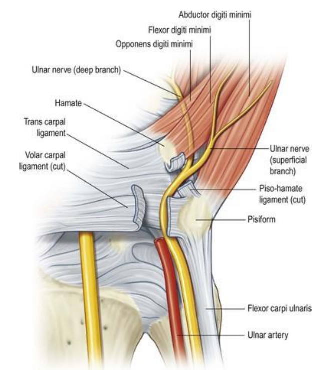




Ulnar nerve in the hand:

The ***Canal of* Guyon** is formed by the pisiform, hook of the hamate, superficial extension of the flexor retinaculum, and the pisohamate ligament, which runs across these two bones to form the roof of the canal. (You don't need to worry about the name of this ligament).

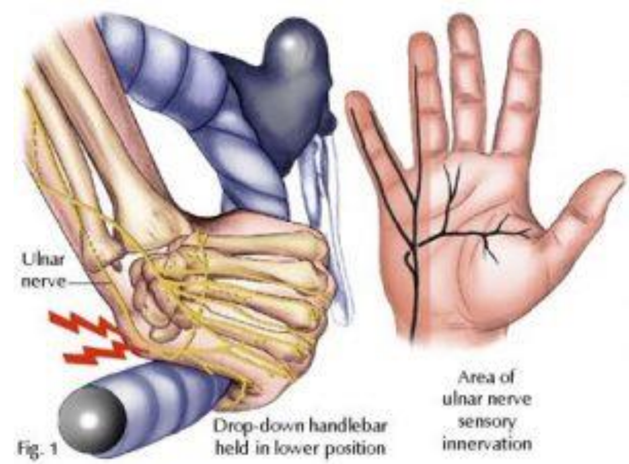




Note this is a LEFT wrist

Guyon's Canal is an important site of ulnar nerve entrapment in the hand.

- Common in cyclists who compress the medial side of the hand by leaning on the handlebars with the weight of their upper body
- Also common with repeated improper hand position in using a computer mouse



http://pelotonphysiotherapy.co.uk/cyclists-palsy-ulnar-nerve-neuropathy/

Balance Health

To determine the functional deficits / symptoms from compression of ulnar nerve in the forearm or hand:

- What is the location of the injury?
- What would be the sensory symptoms?
- What muscles are supplied by the ulnar nerve?
- What are the actions of these muscles?
- With severe weakness or paralysis of these muscles, what actions would be weakened or no longer possible?
- What actions by other muscles that are unaffected would be unopposed?
- What happens to muscles over time when they are weak or paralyzed?

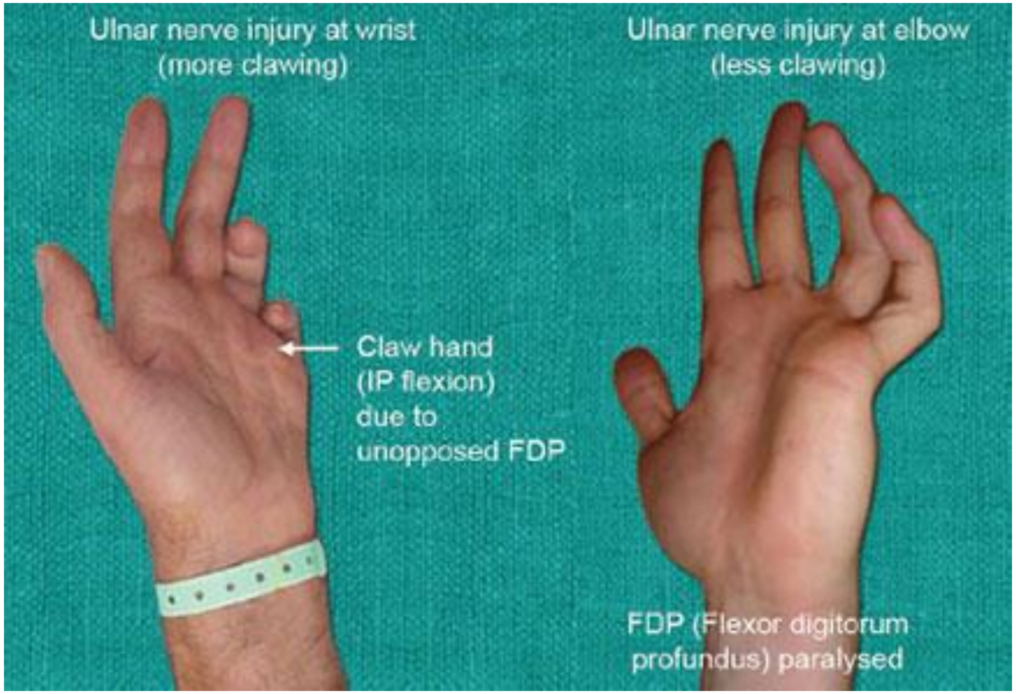
In other words, step through major *sensory* and *motor* symptoms …….don't just assume "claw hand"!

Injury to the ulnar nerve at the wrist:

– Sensory deficits on the medial third of distal palm and medial 1.5 digits
– Patient cannot abduct or adduct the fingers and cannot adduct the thumb
– Patient cannot actively extend the IP joints due to loss of interossei and most of lumbricals
– This leads to an imbalance at the MCP joint – as EDC can still extend, but the intrinsics cannot do their normal flexion at MCP, so get hyperextension at MCP but unopposed flexion at IP joints (due to FDS and FDP) – net effect is a "clawing" of the medial two digits

Injury to the ulnar nerve at the elbow (very similar to injury at wrist):

- Sensory deficits on the medial third of both the palm and the dorsum of hand, plus medial 1.5 digits
- Flexion of wrist yields abduction (as FCR is unopposed)
- Patient cannot abduct or adduct the fingers and cannot adduct the thumb
- Patient cannot actively extend the IP joints due to loss of interossei and most of lumbricals
- This still leads to an imbalance at the MCP joint – as EDC can still extend, but the intrinsics cannot do their normal flexion at MCP, so get hyperextension at MCP but since the medial FDP isn't functioning, get less "clawing" of the medial two digits – i.e., only that caused by FDS



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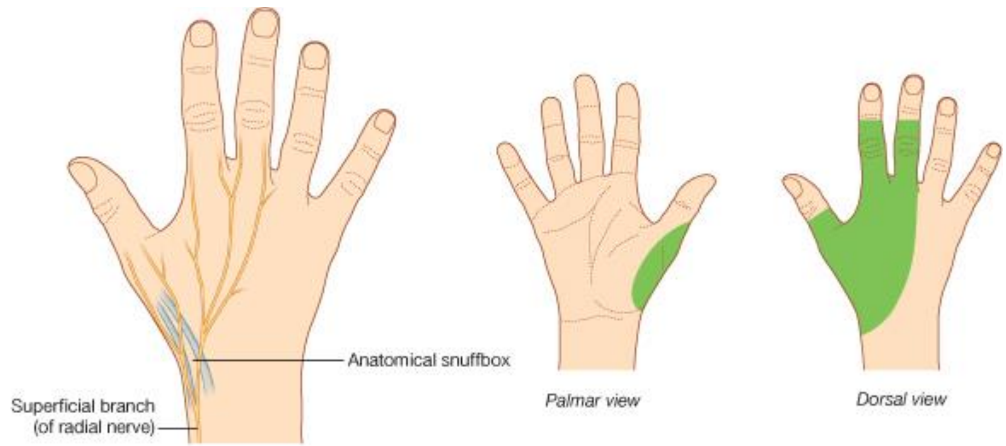
Radial nerve in the hand:

The superficial radial nerve runs along the dorsolateral aspect of the wrist, superficial to the anatomical snuffbox, and then enters the hand.

This branch of radial nerve is entirely sensory to skin.

- As there are *no intrinsic muscles of the dorsum of the hand*, there is no motor innervation to the hand by way of radial nerve.



Superficial radial nerve provides cutaneous innervation to a small strip of skin on the palmar radial side of the hand, and the dorsum of the thumb and the 2 lateral fingers (minus the tips).